Congruences for compositional power series

Dev Goyal

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Contribution and review status. ChatGPT (OpenAI) developed the mathematical arguments and the accompanying code autonomously in response to prompts from Dev Goyal. The author initiated the investigation and requested checks, extensions, and formalization. Independent human review of this new work is pending. Dev Goyal is the sole human author. He would bear responsibility for any final release; this draft awaits his review. The formalization package records its own checked coverage; the written proofs below do not assume that every statement has been verified by a proof assistant.

Abstract

For each positive integer r, let $F_r(x) = x + O(x^2)$ be the formal power series determined by $F_r(x) = x + F_r^{\circ r}(x) F_r^{\circ(r+1)}(x)$, where the superscript $\circ k$ denotes composition k times. We determine every iterate modulo $2r$. For odd r the reduction of F_r is $x/(1-x)$; for even r it is $x/(1-x) + rx^4/(1-x)^3$. This proves the four congruence assertions recorded for OEIS A396099. It also proves seven iterate assertions for A396798, corrects the starting index of a residue block in that entry, and proves the two assertions for A396807. The $r = 3$ case was proved previously by Sundaram. For a second family, defined by $G_q(x - xG_q(qx)/q) = x$, we prove a uniform congruence modulo $q + 1$ and an exact parity criterion. These give six assertions recorded in A393856–A393859. The arguments use coefficientwise uniqueness, an elementary substitution identity, and rational comparison series. All required formal-series facts are proved below.

1 The question and the main result

The sequence OEIS A396099 begins

$$1,\ 1,\ 5,\ 39,\ 383,\ 4357,\ 55057,\ 754223, \ldots.$$

Paul D. Hanna defined its generating function $A(x) = \sum_{n \geq 1} a_n x^n$ by

$$A(x) = x + A(A(x))\, A(A(A(x))). \tag{1}$$

The large coefficients have a simple apparent pattern: they are odd, and their remainders modulo four repeat $1, 3, 3, 1$ starting with the third coefficient. The entry also records two assertions involving compositions of A [1].

We write $U^{\circ k}$ for a compositional iterate, so that $U^{\circ 2}(x) = U(U(x))$. Ordinary powers, such as $U(x)^2$, mean multiplication. Our three basic identities are

$$A(x) \equiv \frac{x}{1-x} + \frac{2x^4}{(1-x)^3} \pmod 4, \tag{2}$$

$$A^{\circ 2}(x) \equiv x + 2x^2 \pmod 4, \tag{3}$$

$$A(x) - xA^{\circ 3}(x) \equiv x + \frac{2x^3}{1-x} \pmod 4. \tag{4}$$

A congruence of power series means that the corresponding coefficients have the same remainder. Thus (3) says that every coefficient of the second iterate above degree two is divisible by four. Equation (4) says that every coefficient of the displayed difference above degree two has remainder two. The initial x in this identity is part of the full series statement.

The explanation of (2) is equally explicit. Since

$$\frac{x^4}{(1-x)^3} = x^4 + 3x^5 + 6x^6 + 10x^7 + 15x^8 + \cdots ,$$

we obtain

$$a_n \equiv 1 + 2\binom{n-2}{2} \pmod 4 \qquad (n \geq 2), \tag{5}$$

where $\binom{0}{2} = \binom{1}{2} = 0$. The binomial coefficient is odd precisely at $n \equiv 0, 1 \pmod 4$ in this range. The residues of a_n, from $n = 3$ onward, are therefore $1, 3, 3, 1$ repeated. Every positive-index coefficient is odd.

These facts arise from a whole family. For $r \geq 1$, let F_r satisfy

$$F_r = x + F_r^{\circ r} F_r^{\circ (r+1)}, \qquad F_r = x + O(x^2). \tag{6}$$

Existence and uniqueness are proved in Section 2. Theorem 4.1 gives a formula for all nonnegative iterates of F_r modulo $2r$. Section 6 extends it to integer iterates by using the compositional inverse.

Attribution and status. Hanna recorded the instances considered here and their conjectures [1, 2, 3, 4]. Sundaram proved the $r = 3$ case, including all integer iterates, in September 2026 [5]. His comparison with $x/(1-x)$ and uniqueness argument are directly relevant prior work. The odd-r argument below extends that calculation. The even-r case requires the extra correction $rx^4/(1-x)^3$. The entries for $r = 2, 4, 5$ still labelled the statements discussed here as conjectures when checked on 24 October 2026. This describes the checked sources; it does not establish exclusive historical priority for the results.

2 Formal power series and uniqueness

2.1 Definitions

For a commutative ring R with identity, a formal power series is an expression $U(x) = \sum_{n \geq 0} u_n x^n$ with coefficients in R. The notation $R[[x]]$ denotes the set of these series. Equality means equality of every coefficient. No numerical value is assigned to x, so analytic convergence is not involved.

Addition is coefficientwise, and multiplication is defined by

$$[x^n](UV) = \sum_{j=0}^{n} u_j v_{n-j}.$$

When $V(0) = 0$, substitution is defined by $U(V(x)) = \sum_{j \geq 0} u_j V(x)^j$. Every coefficient receives only finitely many contributions. Associativity and the distributive laws then follow coefficient by coefficient from polynomial identities.

A series is *normalized* if its constant coefficient is zero and its coefficient of x is one. Thus it has the form $x + O(x^2)$. The notation $O(x^N)$ denotes a series with zero coefficients in degrees

below N. In particular, $U - V = O(x^N)$ says that U and V agree in those degrees. Products satisfy $O(x^a)O(x^b) = O(x^{a+b})$.

For a normalized U, define $U^{\circ 0} = x$ and $U^{\circ(k+1)} = U \circ U^{\circ k}$ for $k \geq 0$. Each iterate is normalized. If $U - V = O(x^N)$, then

$$U^{\circ k} - V^{\circ k} = O(x^N) \qquad (k \geq 0). \tag{7}$$

Indeed, this follows by induction on k: split a difference of compositions into the change in the outer series and the change in the inner series. Substitution by a normalized series preserves the degree of vanishing, and factoring $P^j - Q^j$ shows that a change of order N in the inner argument creates no term of smaller degree.

Reduction of integer coefficients modulo an integer commutes with these operations, because each coefficient is a finite sum of products. A series with constant coefficient one has a multiplicative inverse: if $W(0) = 0$, then

$$(1 + W)^{-1} = \sum_{j \geq 0} (-W)^j.$$

This is well defined coefficientwise, and successive terms cancel upon multiplication. It justifies every fraction used below, whose denominator has constant coefficient one.

2.2 The canonical solution

Lemma 2.1. *For every commutative ring R with identity and every integer $r \geq 1$, there is exactly one normalized series $F \in R[[x]]$ satisfying*

$$F = x + F^{\circ r} F^{\circ(r+1)}.$$

Proof. For normalized U, set $\Phi_r(U) = x + U^{\circ r} U^{\circ(r+1)}$. This series is normalized. Suppose $U - V = O(x^N)$, with $N \geq 2$. By (7), corresponding iterates differ by $O(x^N)$. Each iterate starts with x, so

$$\begin{aligned} \Phi_r(U) - \Phi_r(V) = {} & (U^{\circ r} - V^{\circ r}) U^{\circ(r+1)} \\ & + V^{\circ r} (U^{\circ(r+1)} - V^{\circ(r+1)}) = O(x^{N+1}). \end{aligned}$$

Thus applying Φ_r improves agreement by one degree.

For existence, start with $U_0 = x$ and define $U_{j+1} = \Phi_r(U_j)$. We have $U_1 - U_0 = O(x^2)$; the preceding calculation gives $U_{j+1} - U_j = O(x^{j+2})$ by induction. Hence every coefficient eventually stops changing. Define F by those stabilized coefficients. Each coefficient of $\Phi_r(U)$ depends on only finitely many coefficients of U, so the limit satisfies $F = \Phi_r(F)$.

For uniqueness, two normalized solutions agree below degree two. Whenever they agree below degree N, the displayed calculation and their defining equations force agreement below degree $N + 1$. Induction gives agreement in every degree. □

The proof uses no division, so it applies to residue rings even when the modulus is composite. In particular it constructs the integer series F_r in (6). Its image under any ring homomorphism is the canonical solution over the target ring, by uniqueness. We can therefore identify its reduction modulo $2r$ by finding any normalized solution of the reduced equation.

The first coefficients are useful for checking the indexing:

$$F_r(x) = x + x^2 + (2r + 1)x^3 + O(x^4). \tag{8}$$

The product on the right side of (6) starts with x^2, so the second coefficient is one. It follows by induction that $F_r^{\circ j} = x + jx^2 + O(x^3)$. Multiplying the iterates for $j = r$ and $j = r + 1$ gives coefficient $2r + 1$ in degree three.

3 Comparison series and a substitution identity

Define, for an integer c,

$$T_c(x) = \frac{x}{1-cx}, \qquad T = T_1, \qquad H(x) = \frac{x^4}{(1-x)^3}. \tag{9}$$

These are integer power series. Direct substitution gives

$$T_c \circ T_d = T_{c+d}. \tag{10}$$

Consequently $T^{\circ k} = T_k$ for every nonnegative integer k. The coefficient of x^n in T_k is k^{n-1} for $n \geq 2$, and its coefficient of x is one.

The formal derivative is defined algebraically by $U'(x) = \sum_{n\geq 1} n u_n x^{n-1}$. It satisfies the product and substitution rules, which can be proved first for monomials and then coefficientwise. We need the following rule over a ring in which a particular element has square zero.

Lemma 3.1. *Let $\varepsilon \in R$ satisfy $\varepsilon^2 = 0$, and let $u, v \in xR[[x]]$. For every $S \in R[[x]]$,*

$$S(u + \varepsilon v) = S(u) + \varepsilon v S'(u).$$

Proof. For every nonnegative integer j, the binomial theorem gives $(u+\varepsilon v)^j = u^j + j\varepsilon v u^{j-1}$, with the second term interpreted as zero for $j = 0$. Terms containing two or more factors of ε vanish. Multiply by the coefficient of x^j in S and sum. All substitutions are well defined because their inner arguments have zero constant coefficient. □

There is no division by a factorial in this rule. It therefore works in composite-modulus rings.

Lemma 3.2. *The following identities hold in $\mathbb{Z}[[x]]$:*

$$H(x)T'(T(x)) + H(T(x)) = \frac{2x^4}{(1-2x)^3}, \tag{11}$$

$$(1-x)H(x) - x^2 T(x) - x^3 T'(x) = -\frac{2x^3}{1-x}. \tag{12}$$

Proof. We have $T'(x) = (1-x)^{-2}$, so

$$H(x)T'(T(x)) = \frac{x^4}{(1-x)(1-2x)^2}, \qquad H(T(x)) = \frac{x^4}{(1-x)(1-2x)^3}.$$

Their sum has numerator $x^4((1-2x)+1) = 2x^4(1-x)$ over the common denominator $(1-x)(1-2x)^3$. This proves (11). For the second identity, use denominator $(1-x)^2$ to obtain the numerator

$$x^4 - x^3(1-x) - x^3 = -2x^3(1-x).$$

Cancellation gives (12). □

Finally, expanding three geometric series shows that

$$[x^n]H = \begin{cases} 0, & n < 4, \\ \binom{n-2}{2}, & n \geq 4. \end{cases} \tag{13}$$

The coefficient of x^d in $(1-x)^{-3}$ counts triples of nonnegative integers with sum d. For each first entry i there are $d-i+1$ choices for the other two, giving $(d+1)(d+2)/2 = \binom{d+2}{2}$. This proves (13) without an analytic expansion.

4 The general congruence

Theorem 4.1. *Let $r \geq 1$ and $k \geq 0$ be integers. Then*

$$F_r^{\circ k}(x) \equiv \begin{cases} \dfrac{x}{1-kx}, & r \text{ odd or } k \text{ even}, \\ \dfrac{x}{1-kx} + \dfrac{rx^4}{(1-x)^3}, & r \text{ even and } k \text{ odd}, \end{cases} \pmod{2r}. \tag{14}$$

Proof. First suppose r is odd. Since $r+1$ is even,

$$(1-rx)(1-(r+1)x) = 1-(2r+1)x+r(r+1)x^2 \equiv 1-x \pmod{2r}.$$

Using (10), we obtain

$$x + T^{\circ r} T^{\circ (r+1)} \equiv x + \frac{x^2}{1-x} = T \pmod{2r}.$$

Thus T is the normalized solution of the reduced equation. Lemma 2.1 gives $F_r \equiv T \pmod{2r}$, and composition yields $F_r^{\circ k} \equiv T_k$ for all $k \geq 0$.

Now suppose r is even. Work in the coefficient ring $R = \mathbb{Z}/(2r)\mathbb{Z}$, and set $f = T + rH$. The image of r in this ring has square zero, because r^2 is a multiple of $2r$. Lemma 3.1 and (11) give

$$\begin{aligned} f^{\circ 2} &= T(T+rH) + rH(T+rH) \\ &= T^{\circ 2} + r\big(HT'(T) + H(T)\big) \\ &= T^{\circ 2} + \frac{2rx^4}{(1-2x)^3} = T^{\circ 2} \quad \text{in } R[[x]]. \end{aligned}$$

Because r is even, repeated composition of this identity gives

$$f^{\circ r} = T^{\circ r} = \frac{x}{1-rx} = x + rx^2 \quad \text{in } R[[x]].$$

The last equality again uses $r^2 = 0$ in R. Hence

$$f^{\circ (r+1)} = f(x+rx^2) = f + rx^2 f'.$$

We can now verify the defining equation. After expanding and discarding terms with a factor of r^2, its residual is

$$\begin{aligned} f - \big(x + f^{\circ r} f^{\circ (r+1)}\big) &= (1-x)f - x - rx^2 f - rx^3 f' \\ &= r\big((1-x)H - x^2 T - x^3 T'\big) \\ &= -\frac{2rx^3}{1-x} = 0 \quad \text{in } R[[x]], \end{aligned}$$

where (12) proves the penultimate equality. By uniqueness, the reduction of F_r is f.

For $k = 2j$, the equality $f^{\circ 2} = T^{\circ 2}$ gives $f^{\circ k} = T^{\circ k} = T_k$. For $k = 2j+1$, it gives

$$f^{\circ k} = f \circ T_{2j} = T_k + rH(T_{2j}).$$

The integer series T_{2j} is congruent to x modulo two. Therefore $H(T_{2j}) - H(x)$ has even integer coefficients, and multiplication by r makes it zero modulo $2r$. Thus $f^{\circ k} = T_k + rH$ in $R[[x]]$, as required. □

Corollary 4.2. *Write $h_n = [x^n]H$ as in (13), and let $\epsilon(r,k)$ equal one if r is even and k is odd, and zero otherwise. For every $r \geq 1$, $k \geq 0$, and $n \geq 2$,*

$$[x^n]F_r^{\circ k} \equiv k^{n-1} + r\epsilon(r,k)h_n \pmod{2r}.$$

The coefficient of degree zero is zero, and that of degree one is one. Every positive-index coefficient of F_r is odd.

Proof. The coefficient formula follows from the geometric series for T_k and (13). At $k = 1$, the correction is absent for odd r and is a multiple of the even integer r for even r. Since 2 divides $2r$, all positive-index coefficients are odd. □

5 The four assertions for A396099

Take $r = 2$, so $A = F_2$. The cases $k = 1$ and $k = 2$ of Theorem 4.1 immediately give (2) and (3). The parity assertion follows from Corollary 4.2.

For completeness, the phase of the residue block can be checked directly. For $n \geq 2$, put $b_n = \binom{n-2}{2}$, with value zero when $n = 2, 3$. Then

$$b_{n+4} - b_n = 4n - 2,$$

which is even. The initial values at $n = 2, 3, 4, 5$ are $0, 0, 1, 3$. Thus b_n is even at $n \equiv 2, 3 \pmod 4$ and odd at $n \equiv 0, 1 \pmod 4$. Equation (5) gives

$$a_{4j+3} \equiv 1, \quad a_{4j+4} \equiv 3, \quad a_{4j+5} \equiv 3, \quad a_{4j+6} \equiv 1 \pmod 4 \qquad (j \geq 0).$$

This establishes the recorded residue cycle with its correct start.

It remains to prove (4). In $\mathbb{Z}/4\mathbb{Z}[[x]]$, write $f = T + 2H$. The proof of Theorem 4.1 gave $f^{\circ 2} = x + 2x^2$, so Lemma 3.1 gives

$$f^{\circ 3} = f + 2x^2 f'.$$

Consequently

$$\begin{aligned} f - xf^{\circ 3} &= (1-x)f - 2x^3 f' \\ &= x + 2((1-x)H - x^3 T') \\ &= x + 2\frac{x^4 - x^3}{(1-x)^2} = x - \frac{2x^3}{1-x} = x + \frac{2x^3}{1-x} \pmod 4. \end{aligned}$$

Since $A \equiv f \pmod 4$, this proves (4). Its coefficients above degree two are all congruent to two. All four assertions in A396099 have now been proved.

6 Integer iterates and exact composition order

The statements for A396797 and A396807 include negative iterate indices. We give the extension explicitly.

Lemma 6.1. *Every normalized series over a commutative ring has a unique normalized two-sided inverse under composition.*

Proof. For $U = x + \sum_{n\geq 2} u_n x^n$, seek $V = x + \sum_{n\geq 2} v_n x^n$ with $U(V) = x$. The coefficient equation in degree n has the form v_n plus an expression involving only earlier coefficients of V. Indeed, in V^j for $j \geq 2$, a coefficient of degree n uses only indices at most $n - j + 1 < n$. Thus every v_n is determined without division, giving a unique right inverse. Apply the construction to V to obtain W with $V \circ W = x$. Associativity gives $U = U \circ (V \circ W) = (U \circ V) \circ W = W$. It follows that $V \circ U = x$ too. □

For an integer $k < 0$, define $F_r^{\circ k}$ by composing the inverse $-k$ times. The usual law of integer powers holds in this composition group. Reducing coefficients commutes with the inverse, because it preserves both inverse identities and the normalized inverse is unique.

Corollary 6.2. *The reduction of F_r modulo $2r$ has exact order $2r$ under composition. Formula* (14) *remains valid for every integer k, including negative values. In particular,*

$$F_r^{\circ k} \equiv x \pmod{2r} \quad \Longleftrightarrow \quad 2r \mid k.$$

Proof. For nonnegative k, the coefficient of x^2 in the formula of Theorem 4.1 is k modulo $2r$, because H has no terms below degree four. Equality with x therefore forces $2r \mid k$. Conversely, if $2r \mid k$, then k is even, $T_k \equiv x \pmod{2r}$, and the correction term is absent. This proves the order statement.

For any integer k, choose the unique j with $0 \leq j < 2r$ and $j \equiv k \pmod{2r}$. In the composition group modulo $2r$, the order statement gives $F_r^{\circ k} = F_r^{\circ j}$. Also $T_k \equiv T_j \pmod{2r}$, and k and j have the same parity, because $2r$ is even. The formula for j is therefore exactly the required formula for k. The same coefficient-of-x^2 argument proves the final equivalence for all integer k. □

7 Related OEIS assertions

7.1 A396798: seven iterate assertions and a phase correction

The entry A396798 is the case $r = 4$ [3]. Let $B = F_4$. Theorem 4.1 gives

$$B^{\circ k} \equiv T_k + 4\,\mathbf{1}_{\{k \text{ odd}\}} H \pmod{8}.$$

For even $k = 2, 4, 6, 8$, the geometric series truncates modulo eight as shown in the table below. For odd k, we have $k^2 \equiv 1 \pmod{8}$. Using the parity pattern of h_n, the coefficients from degree two repeat $k, 1, k+4, 5$ modulo eight. These calculations prove all seven iterate assertions recorded in the entry.

Iterate	Reduction modulo eight
$B^{\circ 2}$	$x + 2x^2 + 4x^3$
$B^{\circ 3}$	coefficients $3, 1, 7, 5$ repeated from degree two
$B^{\circ 4}$	$x + 4x^2$
$B^{\circ 5}$	coefficients $5, 1, 1, 5$ repeated from degree two
$B^{\circ 6}$	$x + 6x^2 + 4x^3$
$B^{\circ 7}$	coefficients $7, 1, 3, 5$ repeated from degree two
$B^{\circ 8}$	x

At $k = 1$, the correct coefficient block is $1, 1, 5, 5$ beginning at *degree two*. The entry begins this block at degree one; that phase is false. Equation (8) gives

$$B = x + x^2 + 9x^3 + O(x^4).$$

Thus its third coefficient has remainder one modulo eight, whereas the stated phase would require five. The corrected assertion is

$$[x^{4j+2}]B \equiv 1, \quad [x^{4j+3}]B \equiv 1, \quad [x^{4j+4}]B \equiv 5, \quad [x^{4j+5}]B \equiv 5 \pmod 8 \qquad (j \geq 0).$$

It follows for all indices from the general theorem.

7.2 A396807 and the previously proved A396797 case

For $r = 5$, the series is OEIS A396807 [4]. The odd-parameter case of the theorem, together with Corollary 6.2, gives

$$F_5^{\circ k} \equiv \frac{x}{1 - kx} \pmod{10} \qquad (k \in \mathbb{Z}).$$

Its coefficient of x is one. For $n \geq 2$, its coefficient of x^n is k^{n-1} modulo ten. Taking $k = 1$ gives remainder one for every positive-index coefficient of F_5. These are the two assertions in the entry. Stating degree one separately also handles $k = 0$ without relying on an unexplained expression 0^0.

The same odd-parameter argument at $r = 3$ recovers Sundaram's result for A396797 [2, 5]. We include it as an instance of the general formula, with the prior result explicitly credited.

8 A scaled inverse family

A second family shows how the same uniqueness principle can lead to a different arithmetic pattern. For a positive integer q, consider the normalized series G_q satisfying

$$G_q\left(x - \frac{xG_q(qx)}{q}\right) = x. \tag{15}$$

Hanna's entries A393856, A393857, A393858, and A393859 correspond respectively to $q = 4, 5, 6, 7$ [6, 7, 8, 9]. Each entry proposes that every positive-index coefficient has remainder one modulo $q + 1$. The entries for $q = 4, 6$ also propose that a coefficient is odd exactly when its index is a power of two. These six statements remained labelled conjectures in the sources checked on 24 October 2026.

8.1 Removing the apparent division

If $U(x) = \sum_{n\geq 1} u_n x^n$, define

$$L_q(U)(x) = \sum_{n\geq 1} q^{n-1} u_n x^n.$$

For a nonzero integer q, this is exactly $U(qx)/q$. The coefficient definition makes sense over every commutative ring, without requiring that q be invertible. Equation (15) therefore means

$$G_q\big(x - xL_q(G_q)(x)\big) = x. \tag{16}$$

Lemma 8.1. *For any commutative ring R with identity and any $q \in R$, equation* (16) *has a unique normalized solution in $R[[x]]$. Its construction commutes with ring homomorphisms.*

Proof. Write $G = x + \sum_{n\geq 2} g_n x^n$ and $V = x - xL_q(G)$. The series V is normalized. The coefficient g_n inside V first occurs in degree $n + 1$, because of the initial factor x. Thus the coefficient of degree n in V depends only on $g_2, \ldots, g_{n-1}$.

In $G(V) = V + \sum_{j\geq 2} g_j V^j$, the term $g_n V^n$ contributes exactly g_n in degree n. The terms with $j < n$ depend only on earlier coefficients, and those with $j > n$ contribute nothing in that degree. Therefore the equation in degree n has the form $g_n + P_n = 0$, with P_n determined by earlier coefficients and q. This gives existence and uniqueness by recursion, without division. It also shows that applying a ring homomorphism to the coefficients preserves the equation; uniqueness identifies the resulting solution. □

8.2 The congruence and parity theorems

Theorem 8.2. *For every positive integer q,*

$$G_q(x) \equiv \frac{x}{1-x} \pmod{q+1}.$$

In particular, every coefficient of positive degree is congruent to one modulo $q+1$.

Proof. Work in $R = \mathbb{Z}/(q+1)\mathbb{Z}$, where $q = -1$. For $T = x/(1-x)$,

$$L_{-1}(T) = x - x^2 + x^3 - \cdots = \frac{x}{1+x}.$$

Hence

$$x - xL_{-1}(T) = x - \frac{x^2}{1+x} = \frac{x}{1+x},$$

and direct substitution gives

$$T\left(\frac{x}{1+x}\right) = \frac{x/(1+x)}{1-x/(1+x)} = x.$$

Thus T is the normalized solution of the reduced equation. Lemma 8.1 identifies it with the reduction of G_q. This argument also covers composite moduli such as six and eight. □

Theorem 8.3. *If q is a positive even integer, then*

$$G_q(x) \equiv \sum_{j\geq 0} x^{2^j} \pmod{2}.$$

Equivalently, its coefficient of degree $n \geq 0$ is odd if and only if n is a power of two. If q is odd, all positive-index coefficients of G_q are odd.

Proof. For even q, reduction modulo two gives $L_q(U) = x$ for every normalized U: all factors q^{n-1} with $n \geq 2$ vanish. Therefore the reduced equation is

$$G_q(x + x^2) = x.$$

Let $S(x) = \sum_{j\geq 0} x^{2^j}$ over $\mathbb{Z}/2\mathbb{Z}$. The strictly increasing exponents make this a well-defined formal series. Repeated squaring in characteristic two gives $(x+x^2)^{2^j} = x^{2^j} + x^{2^{j+1}}$, and hence

$$S(x + x^2) = \sum_{j\geq 0} \left(x^{2^j} + x^{2^{j+1}}\right) = x.$$

Every term beyond the first occurs twice and cancels. By uniqueness, the reduction of G_q is S. Its support is exactly the powers of two, including $1 = 2^0$ and excluding zero.

If q is odd, then $q+1$ is even, so Theorem 8.2 already gives oddness of every positive-index coefficient. □

The four congruence applications and two parity applications are summarized below. They all concern the solution of the original equation (15), not a separately defined sequence chosen to fit the remainders.

q	OEIS entry	Consequences
4	A393856	remainder one modulo five; power-of-two parity
5	A393857	remainder one modulo six
6	A393858	remainder one modulo seven; power-of-two parity
7	A393859	remainder one modulo eight

Verification and responsibility

The proofs in this article apply to all indices. Exact finite computations in the accompanying package reconstruct coefficients from the defining equation and check the reductions, compositions, and displayed examples. Such computations can expose indexing or implementation errors; they are separate from the all-index proofs.

The accompanying Lean development proves the four original A396099 assertions for the canonical integer series. It also proves the general congruence for every positive parameter and every natural or integer iterate, including the exact composition order; the seven iterate assertions for A396798, its corrected coefficient phase and a counterexample to the stated phase; both A396807 assertions, including negative iterates; and all six assertions for A393856–A393859. The definitions use actual formal power series, and negative iterates use their compositional inverses.

The final package's coverage table, explicit theorem types, fresh build records, and transitive axiom audit are the authoritative record of formal verification. The Lean proofs sometimes take a different algebraic route from the exposition: in particular, exact ring identities replace the derivative calculation in the even-parameter argument. This does not claim a separate formalization of every intermediate lemma in the article. Finite coefficient checks remain corroborating computations, not all-index proofs. Historical priority, source status, and authorship are outside Lean's certification.

ChatGPT (OpenAI) developed the mathematical arguments and the accompanying computational and formalization work autonomously from the author's prompts. The author initiated the project and requested verification, extensions, and revisions; the author did not supply the proof strategy or intermediate mathematical arguments. Independent human review of this new work remains pending. Dev Goyal is the sole human author. The intended responsibility for a final release rests with him, including its mathematical claims, references, attribution, and any errors; this draft awaits his review. The verification status of this work is independent of any earlier project by the author.

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